

**Use of a Single Source Below the Simplified Acquisition Threshold (SAT)****Section A: General Contract Information**Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)?☐ Yes☒ No

Contracting Activity: AFRL/PZLET

Project/Program Name: AdaCore - GNAT Pro/SPARK Pro/GNAT DAS

Purchase Request / Local ID Number: [REDACTED]

Estimated Contract Cost (including Options): [REDACTED]

Section B: Description of the Supplies/Services Required

The Government has a requirement for a Sole Source purchase from AdaCore Technology, Software license and support for one calendar year of GNAT Pro, GNAT DAS and SPARK Pro, a code verification software specific to one manufacturer.

This acquisition will utilize FY26 3600 Funds and will be for a 1-year License Period of 31 May 2026 through 30 May 2027

Section C: Rationale for Soliciting from a Single Source or Brand Name

The need for rigorously or formally verified complex autonomy software has been emphasized in a number of strategic planning documents produced by the USAF and DoD, including "Technology Horizons," "Autonomous Horizons: The Way Forward," and the "DoD Digital Engineering Strategy." AFRL/RQQA performs R&D of autonomy-related software for unmanned aerial vehicles. To meet the needs of the USAF, we need tools to formally verify this software.

AFRL/RQQA ran a pilot program to solicit solutions for formal software verification. Out of all the proposed solutions, SPARK Pro was the only product that could feasibly be used to verify our software. The combination of a language designed for safety, integrated contract-based specifications, and a high degree of automation in its proof tools creates a powerful environment for building high-integrity software. For many safety-critical applications, having qualified tools and certification support is a key requirement. AdaCore provides certification and qualification material for standards like DO-178C, ISO 26262, and EN 50128, which can be a significant advantage over tools that lack this level of commercial and certification support. Competing products were not able to verify necessary properties of our software with reasonable amounts of effort such as the certification for appropriate use in standards or the consolidation of workflows into a single environment that maximizes efficiency in going from specification to proven implementation. Therefore, AFRL/RQQA now uses the SPARK language and SPARK Pro tools to write and formally verify our software.

AdaCore is the sole source of GNAT Pro Enterprise, GNAT Dynamic Analysis Suite (DAS), and SPARK Pro. GNAT Pro Enterprise is a complete development environment for producing critical software systems built in Ada/SPARK, C, and C++ and enables use of SPARK Pro. GNAT DAS is a comprehensive testing solution for software that integrates automated unit testing, fuzzing, and code coverage. SPARK Pro is a collection of tools that perform formal verification of code written in the SPARK subset of Ada.

A large amount of work produced in RQQA relies heavily on the use of Ada/SPARK and the GNAT Pro Enterprise, GNAT DAS and SPARK Pro tool chain. No other tools support the functionality offered by GNAT Pro Enterprise, GNAT DAS, and SPARK Pro for the Ada/Spark programming languages. Switching to a different programming language would entail duplication of approximately 1 years of effort to get back to current status with the caveat that other programming languages do not have equivalent tooling to address the specific verification and validation concerns in a single source toolchain. Attempting to assemble similar capabilities to support a different programming language is currently not possible. The cost associated with developing the capabilities for a different programming language would be equivalent to a team of [REDACTED] working for 1 [REDACTED] to mimic the development effort produce by AdaCore to develop these capabilities in GNAT Pro, GNAT DAS, and SPARK Pro. Additionally, [REDACTED] would need to work for 1 years to replicate the work that has been performed by the RQQA team, or equivalent to approximately [REDACTED] in duplication of cost that would not be recovered.

Section D: Efforts to Obtain Competition



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Since adopting the use of GNAT Pro, GNAT DAS, and SPARK pro, AFRL/RQQA has annually monitored the solution space through participation in conferences focused on formal software verification solutions, literature reviews, and consultations with other government groups that have expertise in formal software verification, including NASA Langley Research Center and NASA Ames Research Center. Based on our findings, SPARK Pro is still the only product that can feasibly be used to verify our software.

Section E: Steps to Preclude Future Single Source or Brand Name Awards

Market research (through consultation with formal software verification experts in industry, academia, and other government agencies) found no other sources that provide tools to perform formal verification of Ada or SPARK code. AFRL/RQ will continue to monitor the market to determine if any other brand names are able to supply the same software functionality.

Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(1\)](#), that the circumstances of this contract action deem only a single source is reasonably available.

Date	Typed Name and Rank/Grade of Contracting Officer	Signature of Contracting Officer
10-Apr-2026	[REDACTED]	[REDACTED]